Atmosic Remote Control Functional Tester

User Guide

SUMMARY: This document describes the setup and usage of the Atmosic Remote Control Functional tester, which provides functional test capability in production environments for ATM2, ATM3 or ATM33/e based remote controls.

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1. Overview

This document describes the features and usage of the Remote Control Functional Tester (RCFT). The RCFT integrates an ATM2 device to support functional testing of an Atmosic based remote control on the production line. The RCFT is designed with an ATM2 series Bluetooth LE device, FTDI 4232, and a printed inverted F-type antenna to support customer product TX/RX functionality verification. The RCFT is used with the Atmosic Production Tool to further support customer mass production in the factory.

2. Remote Control Functional Tester Kit

Figure 1 shows the full unit of the RCFT kit: RCFT with gray enclosure and a USB type-B cable for power and control of the RCFT.

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Figure 1 - Remote Control Functional Tester Kit

2.1 Remote Control Functional Tester

The I/O panel of the RCFT is shown in Figure 2. The USB type-B connector is used to control RCFT by PC/Laptop and provides the main DC power 5 V voltage for RCFT hardware systems. The red color LED is right next to the USB type-B connector is used to indicate RCFT power status, and the system reset button is left next to the USB type-B to force a system reset.



Figure 2 – I/O Panel of Remote Control Functional Tester

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Figure 3 shows two IR LEDs exposed on the front panel of RCFT to support IR RX and IR TX testing respectively. The black LED is an IR receiver and the other transparent LED is the IR Transmitter.



Figure 3 - Front Panel of Remote Control Functional Tester

The main electrical components of the RCFT are depicted in the block diagram shown in Figure 4.

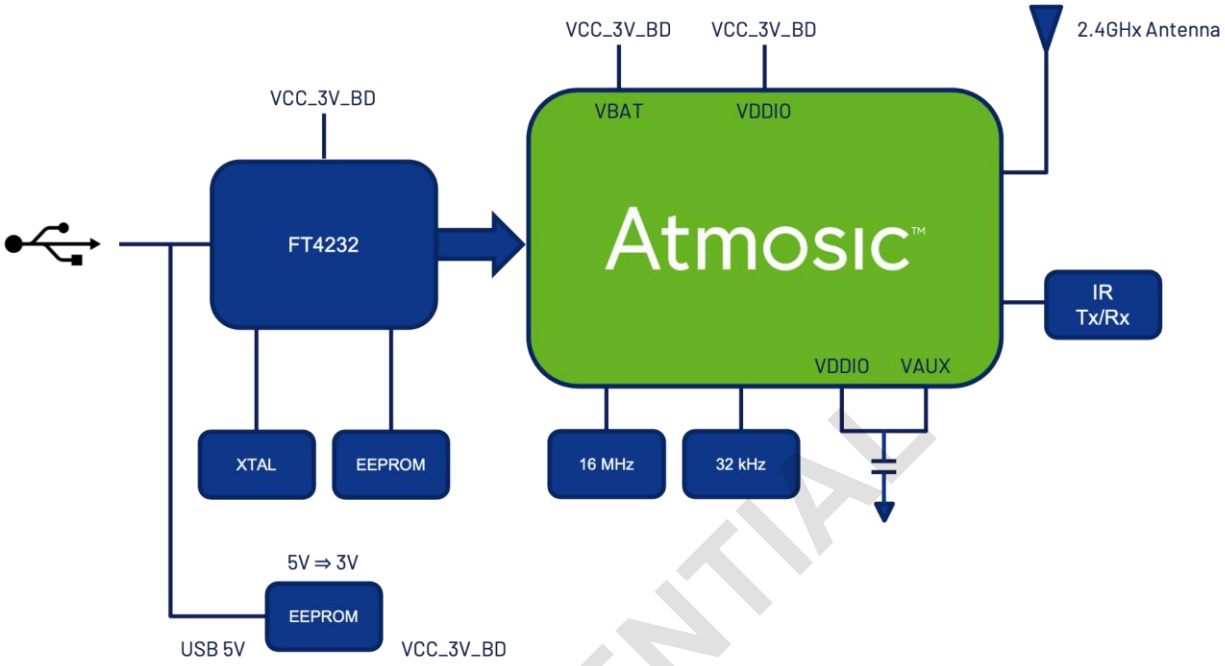


Figure 4 - Remote Control Functional Tester Block Diagram

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3. Remote Control Functional Tester LED

3.1 General Operation

1) Pairing, Re-Pairing and Unpairing:

   There are no external keys/buttons to support RCFT Pairing, Re-Pairing or Un-Pairing with DUT, the pairing/unpairing behavior is controlled by AT commands automatically.

2) RCFT System Reset :

   The system reset button is exposed in the I/O panel and aligned with the USB type-B connector and power LED. The push button is triggered, the system will reset and boot-up automatically.

3.2 Remote Control Functional Tester LED

The related LED function is described below:

1) RCFT power status:

   The RED LED turns on during system power on. Otherwise, the RED LED will be off.

2) IR Receiver LED:

   The IR receiver LED is exposed in the front panel of RCFT for receiving IR signals, the decoded content can be read out on the RCFT console log.

3) IR Transmitter LED:

   The IR transmitter is also exposed in the front panel of RCFT and next to the IR Receiver LED. The function of the IR transmitter is used to send standard IR coded signals to DUT.

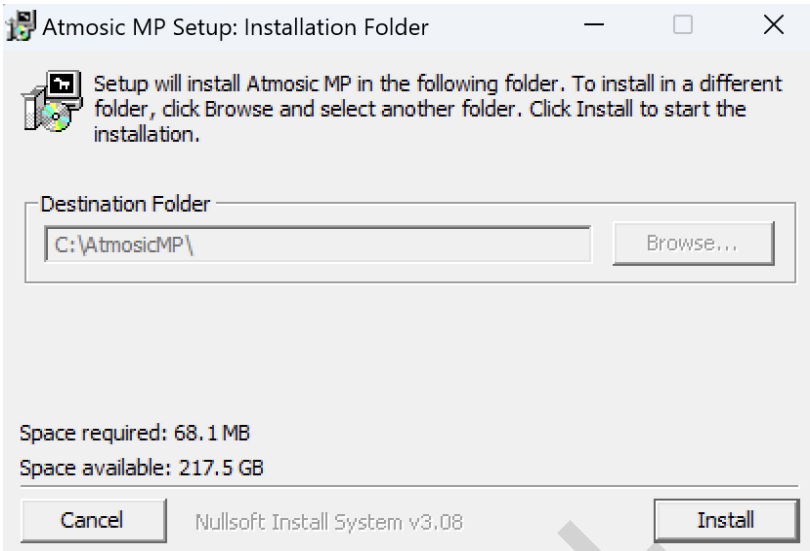
4. Loading New Firmware

A firmware image can be updated by an Atmosic's MP Tool that executes on a PC/Laptop connected to the RCFT via provided USB type-B cable.

The instructions for loading new firmware:

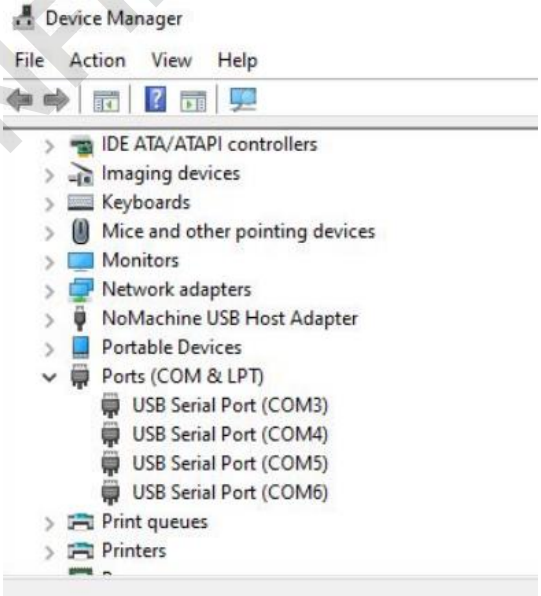
1) Connect the RCFT to the PC/Laptop using the provided USB cable.
2) The Atmosic's MP Tool GUI is provided with the Atmosic Software Development Kit and can be downloaded from the support section of the Atmosic website.
3) Install Atmosic's MP Tool
   a) Installer file name: AtmosicMP_Inst__.zip
   b) e.g. AtmosicMP_Inst_20210804_v2.0.2.0.zip
   c) Install path: C:\AtmosicMP

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4) Check status after installation.
   a) There should be 4 USB serial ports that will appear in “Device Manager” after plug-in the APT/DL board for the first time.

      ***Note**: If 4 USB serial ports do not appear, update the APT/DL board driver.*

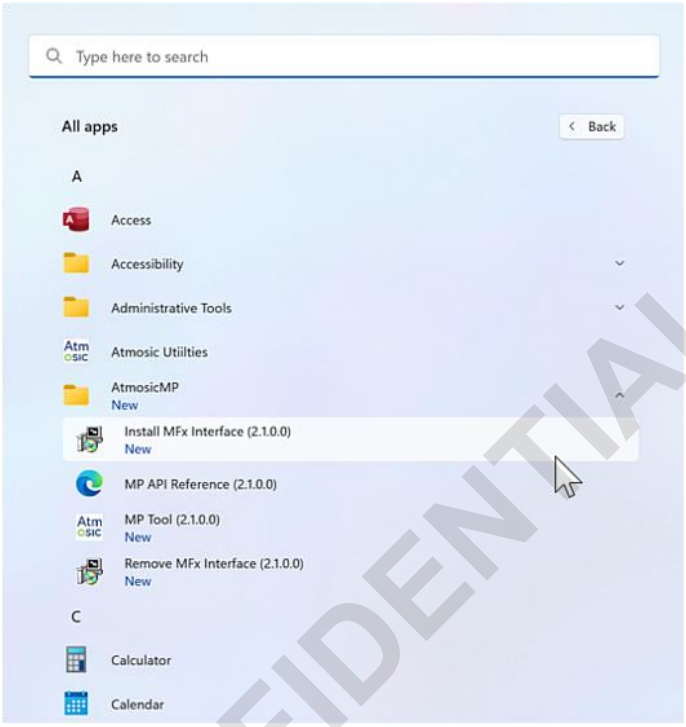


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5) Install MFx interface.

   a) Under Start menu/AtmosicMP (MP Tool installer created)

   b) Click “Install MFx Interface(<Version>)”



   c) Wait for console to auto close

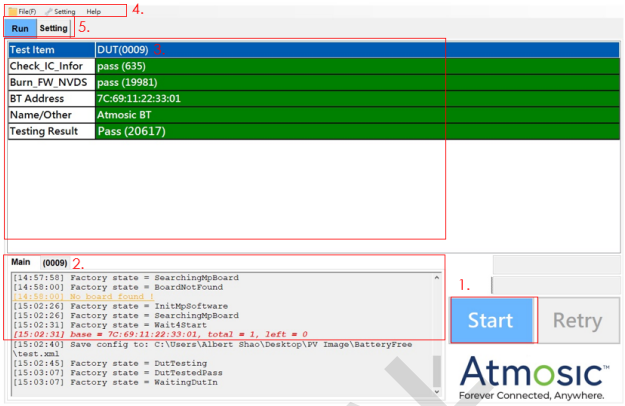
6) After installation of the MFx interface, the first serial port will change to “Atmosic MFG”



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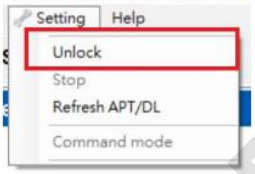
7) After installation of the MFx driver interface for AtmMp tool, execute the MPTool when the following screen appears.

1. Start button
2. Runtime message
3. Test flow result
4. Menu bar
5. Tab
(change main/setting page)



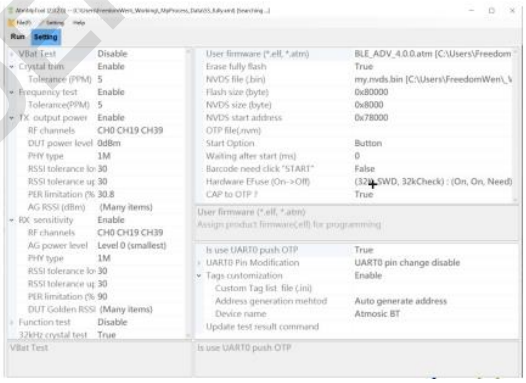
8) Unlock by typing the password “Atmosic123”

- Unlock under menu bar

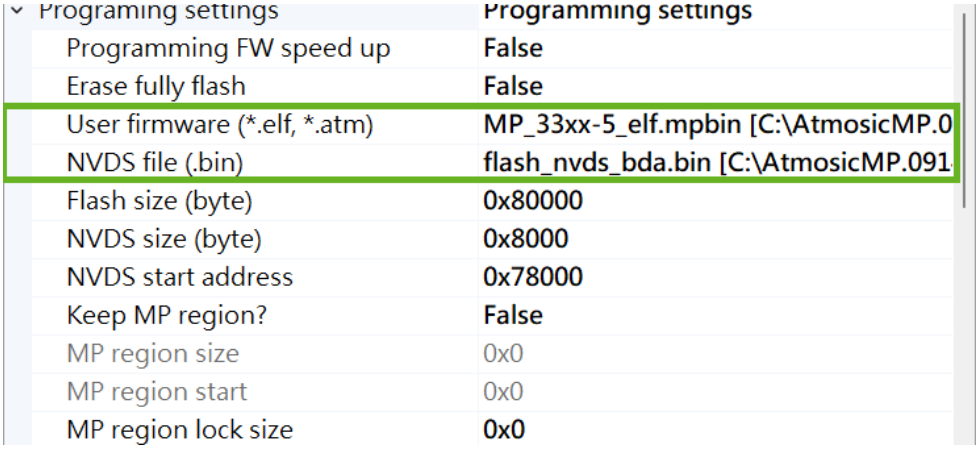


- Password is “Atmosic123”
- Press “enter”





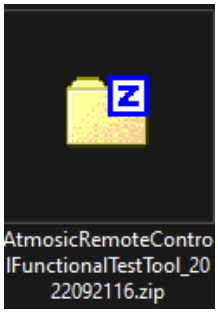
9) Select firmware image to be programmed on “Setting” tab, the detailed as below following shown, then press Start button to program image on “run” tab



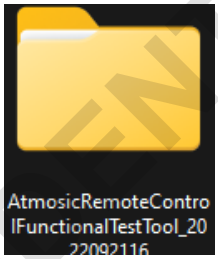
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5. Execute Atmosic Production Tool

The setup package of Atmosic Production Tool is a compressed file named AtmosicProductTool_20xxxxxxxx.zip



1) Extract the AtmosicProductTool_20xxxxxxxx.zip into AtmosicProductTool_20xxxxxxxx



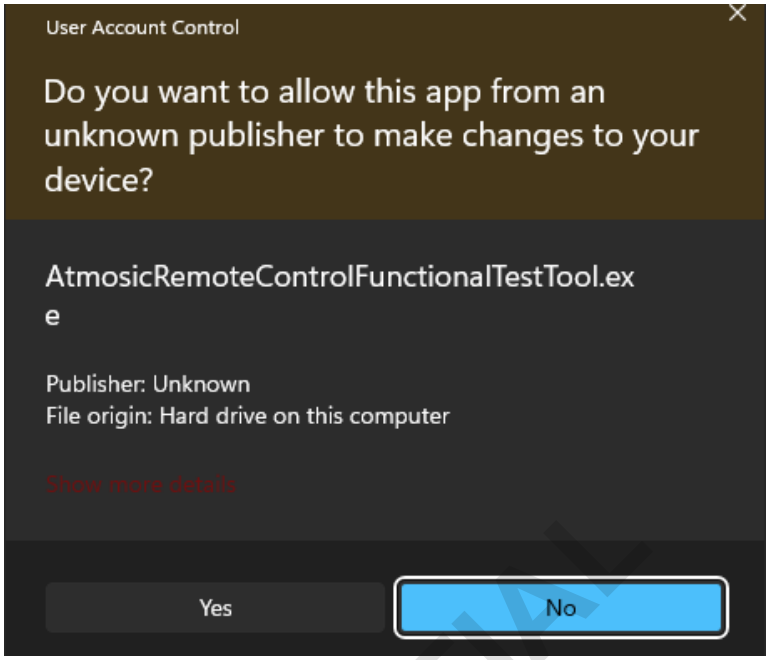
2) Open the folder AtmosicProductTool_20xxxxxxxx

3) Execute AtmosicProductTool.exe

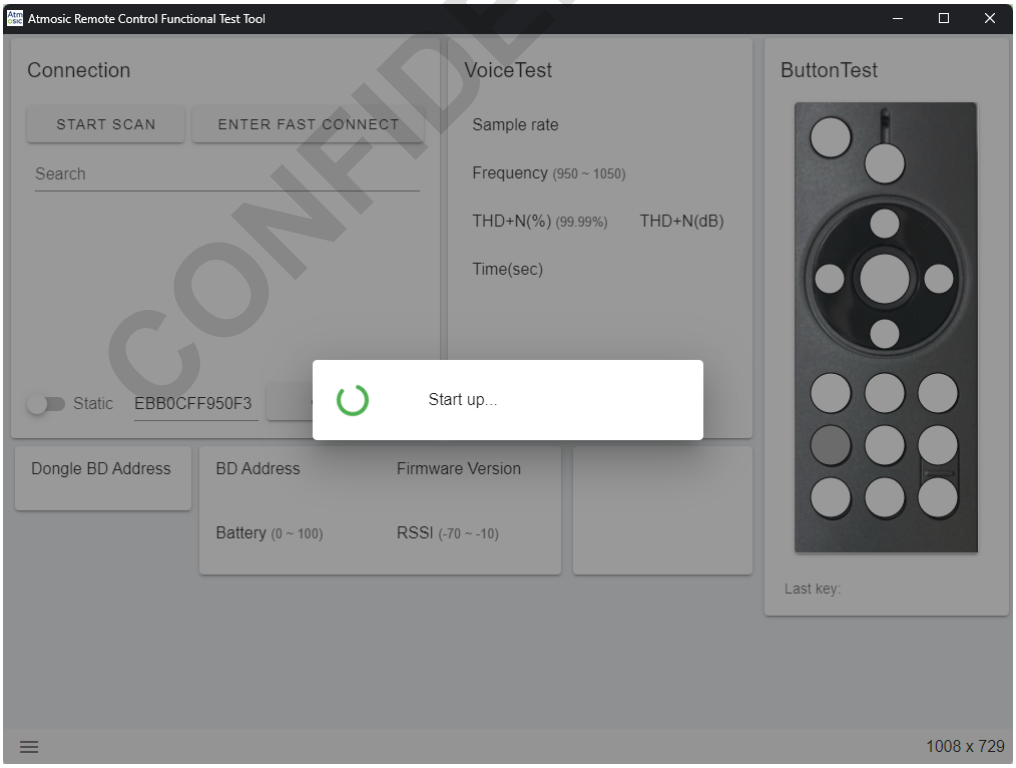


4) Click Yes on the pop-up window

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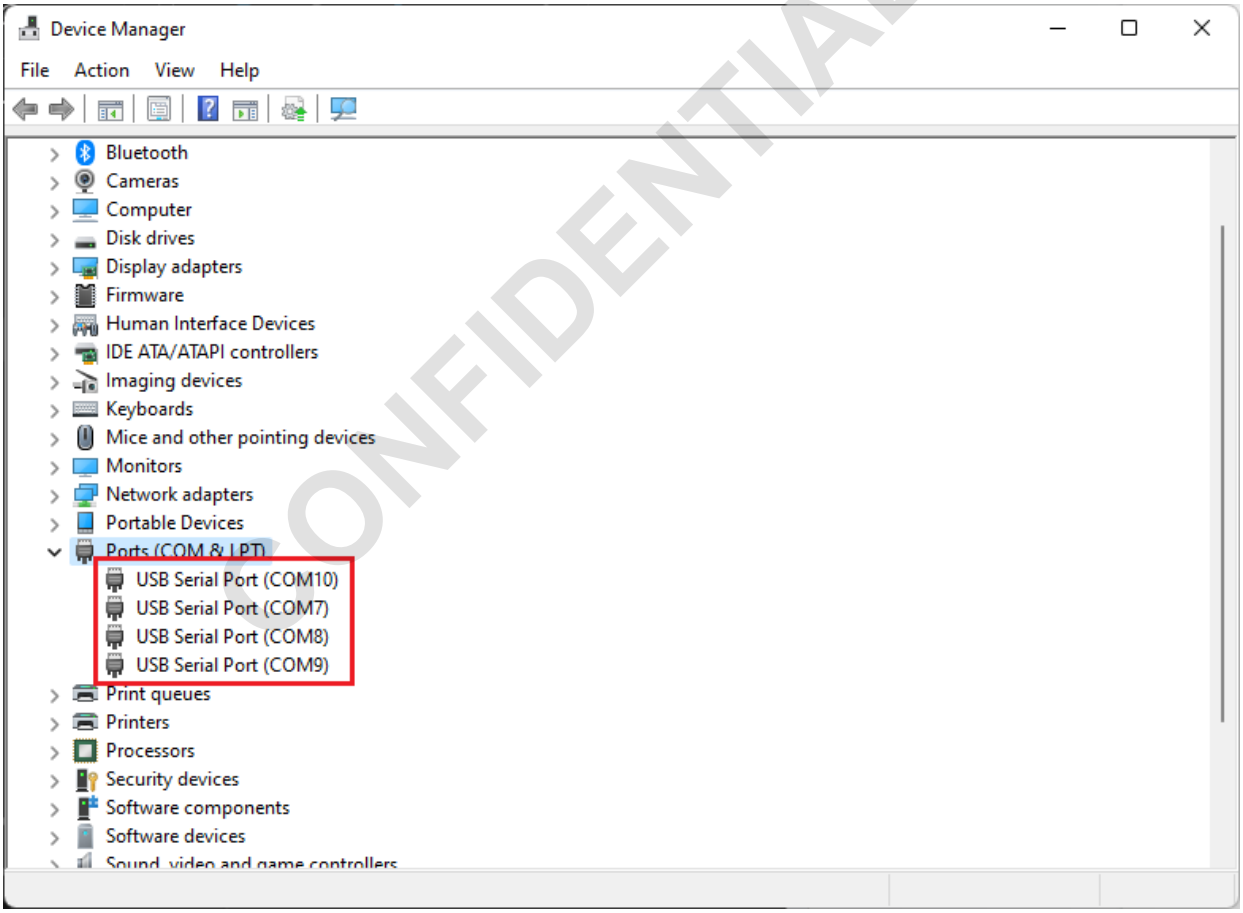
5) Main Window of Atmosic Product Tool is displayed



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5.1 Check COM Port Setting

1) Connect the Atmosic Remote Control Functional Tester to the laptop and wait for ~ 30 seconds) to install the FTDI driver.
2) Open the Windows Device Manager
    a) Open the “Run” dialog box by pressing and holding the Windows key, then press the R key.
    b) Type in “devmgmt.msc” then click the OK button
3) If there are four COM ports listed in the Device Manager after plug-in the Atmosic Remote Control Functional Tester to laptop.



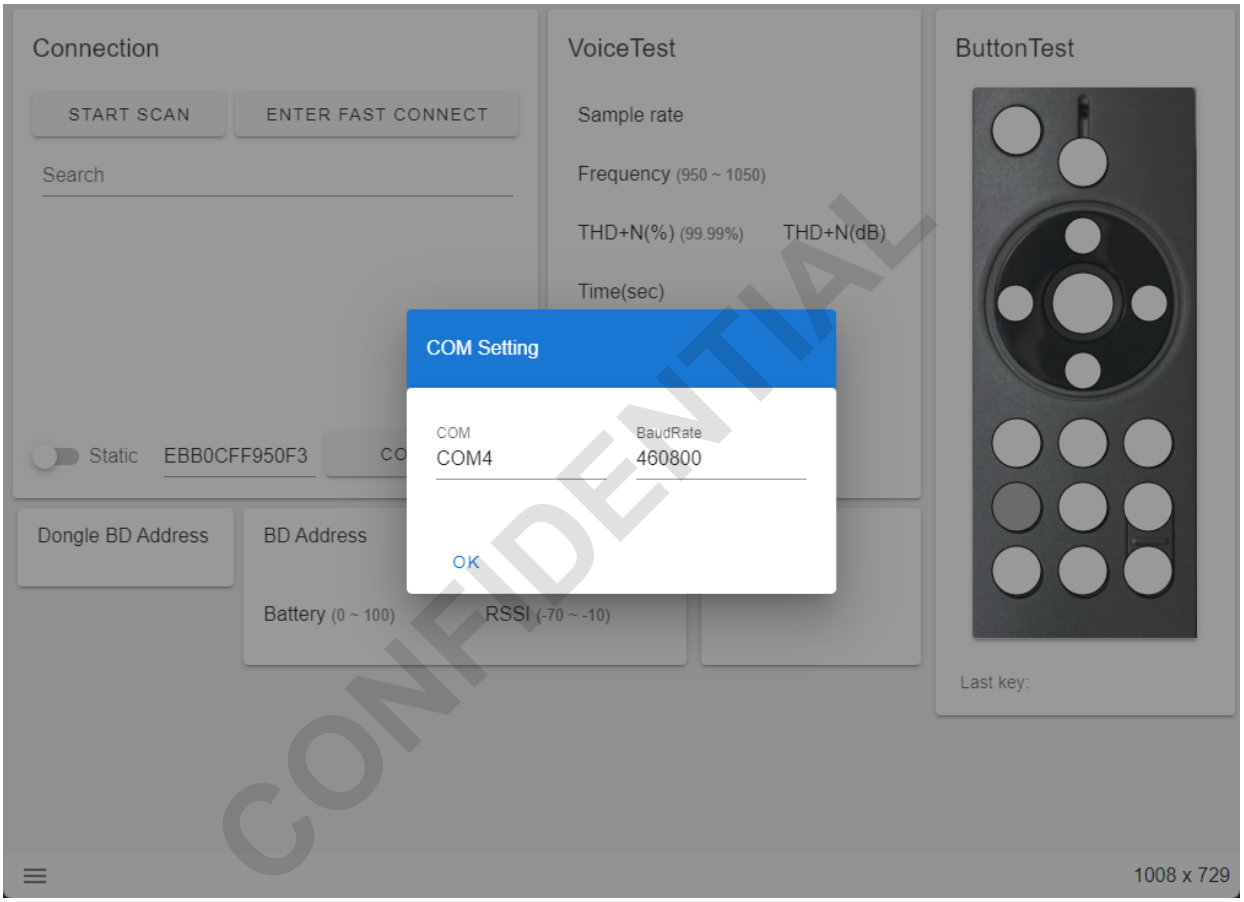
4) If not, please download the FTDI driver and install it before installing the MFx interface driver.
5) In the Atmosic Product Tool, **the third COM port** is the only communication between the tool and the tester.

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6. User Interface

6.1 Landing Page

When the tool is opened, first set the COM port number and baud rate. Refer to Check COM Port Setting for COM port, and below for baud rate.



6.2 Main Page

After setting COM port and baud rate, you can see six blocks in the main page as shown in Figure 6:

1) Connection block: support three methods, for more information, please refer to Connection Modes.
2) Voice test block: shows the result of the Voice test, please refer to Voice Test.
3) Button test block: shows the result about the Button test, please refer to Button Test.
4) Dongle information block: shows some information about Dongle.

5) Remote control information block: shows some information about testing remote control, please refer to Remote Control Information Check.

6) Final test result: there show PASS/FAIL means the testing result.

7) Menu button

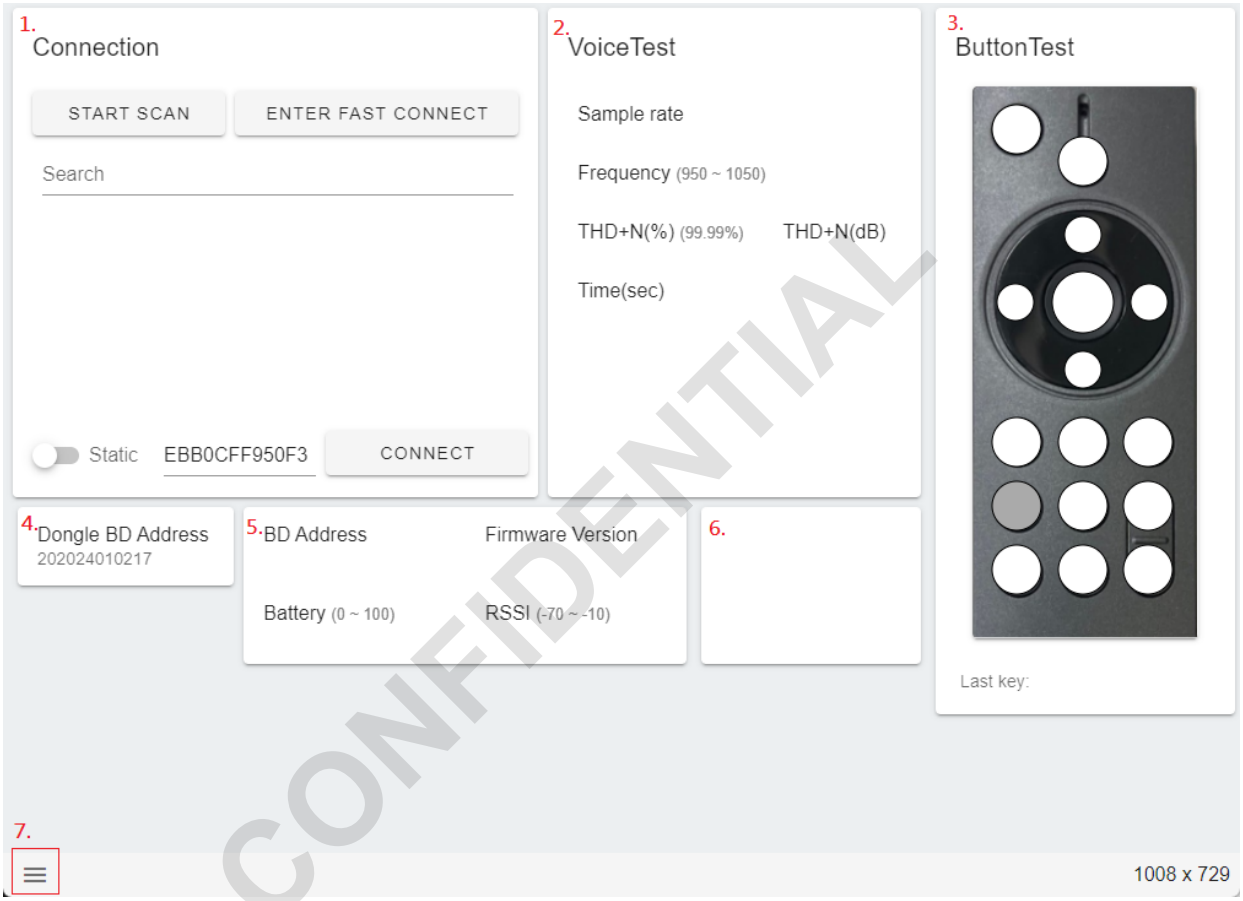


Figure 5 - Main page

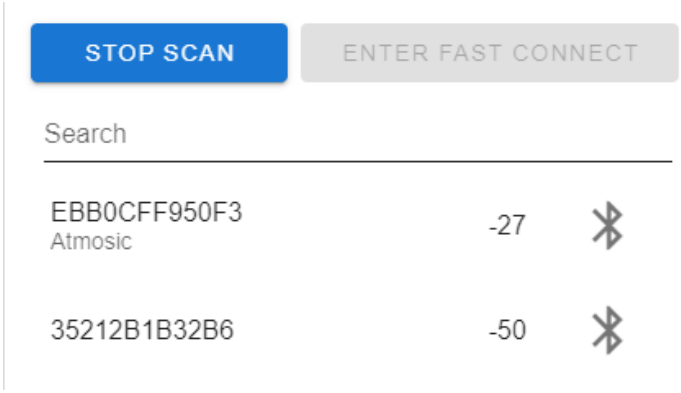
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7. Connection Modes

There are three methods to connect to the remote control.

7.1 Scan Mode

1) Click the “START SCAN” button, and wait for the tester to search nearby Bluetooth devices.



2) Click  button to create connection with the device.

7.2 Barcode Mode

1) Select the option to change Public or Static device (once).
2) Use barcode to scan the Bluetooth Device (BD) address for text input.
3) Click the "CONNECT" button to create a connection with the device.



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7.3 Fast Connect Mode

If your device support fast connect mode, please follow this process

1) Click “ENTER FAST CONNECT”.



2) Press the special key combination on testing remote control.
3) Wait for the connection to establish.

8. Button Test

1) When the button test starts, you can see a countdown timer in the upper left corner of the button test block.
2) When the operator presses a button on the remote, you can see the corresponding button turns green.
3) When the timer expires, all unpressed buttons fail and show red.

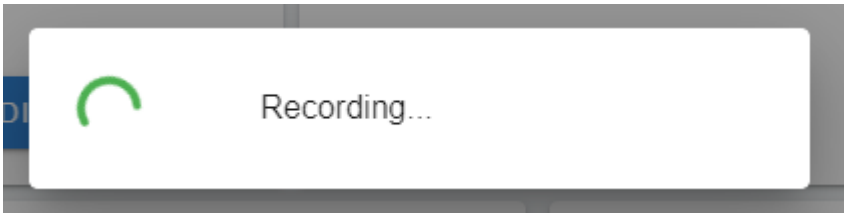


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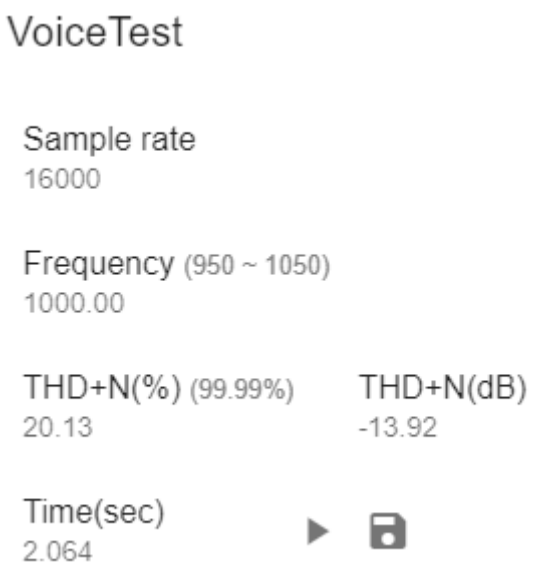
- You can set button layout in the settings file.

9. Voice Test

1) When the voice starts, the dialog shows “Recording…”



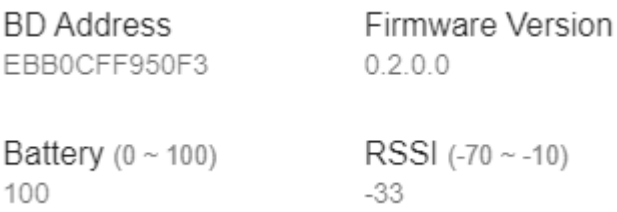
2) When all tests are completed, the test results are displayed in the VoiceTest block.



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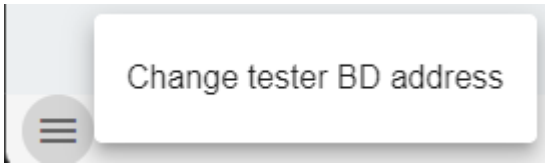
10. Remote Control Information Check

- When the device is connected, the tool will automatically read the following information
  - BD Address
  - Firmware Version
  - Battery Level
  - RSSI



11. Change Tester Bluetooth Device Address

Follow the steps to change the tester's BD address. Click menu button at left-bottom corner and select "Change tester BD address"



1) Key in a new BD address to replace the old address.
2) Click OK to set the new BD address to tester.



12. Settings File

The settings file is located at {ROOT_FOLDER}\html\setting.js.

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```
var setting = {
connection: {
wait_connection_event_time: 1500,
retry_connection_max_times: 5,
hid_handle: [44, 48],
hid_report_handle: [42, 46],
},
button_test: {
enable_test: true,
test_time: 30,
remote_image: "remote_model.png",
button_list: [
{ x: 35, y: 34, r: 20, signal: "HID", value: "42_30" },

{ x: 89, y: 60, r: 20, signal: "HID", value: "42_221" },

{ x: 89, y: 120, r: 15, signal: "HID", value: "42_42" },
{ x: 34, y: 175, r: 15, signal: "HID", value: "42_44" },
{ x: 89, y: 175, r: 25, signal: "HID", value: "42_41" },
{ x: 143, y: 175, r: 15, signal: "HID", value: "42_45" },
{ x: 89, y: 230, r: 15, signal: "HID", value: "42_43" },

{ x: 35, y: 289, r: 20, signal: "HID", value: "42_224" },
{ x: 89, y: 289, r: 20, signal: "HID", value: "42_223" },
{ x: 142, y: 289, r: 20, signal: "HID", value: "46_76" },

{ x: 35, y: 341, r: 20, signal: "IR", value: "NONE" },
{ x: 89, y: 341, r: 20, signal: "HID", value: "42_cd" },
{ x: 142, y: 341, r: 20, signal: "HID", value: "42_e9" },

{ x: 35, y: 393, r: 20, signal: "HID", value: "42_b6" },
{ x: 89, y: 393, r: 20, signal: "HID", value: "42_b5" },
{ x: 142, y: 393, r: 20, signal: "HID", value: "42_ea" },
],
},
voice_test: {
enable_test: true,
record_length: 2,
atvv_version: "0.4",
frequency_high: 1050,
frequency_low: 950,
thdn_limit: 99.99,
},
infors: {
rssi_high: -10,
rssi_low: -70,
get_rssi_max_times: 5,
battery_high: 100,
battery_low: 0,
fw_version: ["0.1.0.0", "0.2.0.0"],
},
};
```

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12.1 Button Test Layout

```
button_test: {
enable_test: true,
test_time: 30,
remote_image: "remote_model.png",
button_list: [
{ x: 35, y: 34, r: 20, signal: "HID", value: "42_30" },

{ x: 89, y: 60, r: 20, signal: "HID", value: "42_221" },

{ x: 89, y: 120, r: 15, signal: "HID", value: "42_42" },
{ x: 34, y: 175, r: 15, signal: "HID", value: "42_44" },
{ x: 89, y: 175, r: 25, signal: "HID", value: "42_41" },
{ x: 143, y: 175, r: 15, signal: "HID", value: "42_45" },
{ x: 89, y: 230, r: 15, signal: "HID", value: "42_43" },

{ x: 35, y: 289, r: 20, signal: "HID", value: "42_224" },
{ x: 89, y: 289, r: 20, signal: "HID", value: "42_223" },
{ x: 142, y: 289, r: 20, signal: "HID", value: "46_76" },

{ x: 35, y: 341, r: 20, signal: "IR", value: "NONE" },
{ x: 89, y: 341, r: 20, signal: "HID", value: "42_cd" },
{ x: 142, y: 341, r: 20, signal: "HID", value: "42_e9" },

{ x: 35, y: 393, r: 20, signal: "HID", value: "42_b6" },
{ x: 89, y: 393, r: 20, signal: "HID", value: "42_b5" },
{ x: 142, y: 393, r: 20, signal: "HID", value: "42_ea" },
],
},
```

12.1.1 Remote control photo

Under the button_test, you can change the remote photo path to “remote_image”

12.1.2 Button setting

You can add the button by this arguments:

- x: The horizontal distance relative to the upper left corner of the remote photo.
- y: The vertical distance relative to the upper left corner of the remote photo.
- signal: HID or IR for the button signal source
- value: Refer the value at ButtonTest block



13. Report File

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	test_time	test_result	bd_addres	fw_version	battery	rssi	button_tou	audio_sam	aduio_freq	audio_thdr	audio_thdr	audio_time_length	
2	20220602_	Fail	EBB0CFF95	0.2.0.0	100	0	HID_42_41	16000	120.12	-1.15	87.65	2.256	
3	20220602_	Pass	EBB0CFF95	0.2.0.0	100	0	HID_42_41	16000	1000	-13.92	20.13	2.064	
4	20220602_	NotComple	EBB0CFF950F3		0	0		0	0	0	0	0	
5	20220602_	Fail	EBB0CFF95	0.2.0.0	100	0		0	0	0	0	0	
6	20220602_	Fail	EBB0CFF95	0.2.0.0	100	0	HID_42_42	16000	1077.54	-0.08	99.13	2.128	
7	20220602_	Pass	EBB0CFF95	0.2.0.0	100	0	HID_42_41	16000	1000	-13.73	20.59	2.112	

The Atmosic product tool has a report file in CSV format and is stored in the logs folder under the root directory of the Tool, named testlog_{date}.csv

14. Tool Detail Log

The detail log is at {ROOT_FOLDER}\logs\.

Use folders to separate and name them {date}_{time} according to the tool startup time.

15. Limitations

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The remote control functional tester as provided does not support the following features:

- Infrared (IR) Tx support - by default, the atmosic remote control functional tester firmware does not transmit any data via IR Tx, the IR Rx is default setting. Support for IR Tx requires a new firmware image. If support for these features is desired, contact your local Atmosic representative for assistance.

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Revision History

Date	Version	Description
September 28, 2022	0.50	Initial version created.

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